

§1

THE SITUATION

Everything that could hurt Singapore arrives from beyond our borders — across the sea approaches, or through a handful of air corridors — into a country fifty kilometres wide. There is no depth here, no rear area and no second chance — warning time is the only defence we have, and it is measured in minutes. The information that would buy those minutes already exists: radars, cameras, transponders, shipping data, flight logs, and the ordinary traffic of one of the world's busiest sea lanes. It sits in a dozen systems that disagree about identity, about time, and about how far each other should be believed, and nobody has joined it. Collection stopped being the problem years ago. We are drowning in signals and blind anyway.

PROBLEM STATEMENT

Ideate a way to fuse sources nobody has joined — sensors, open data, and things that were never meant to be sensors — into usable warning, either that a conflict is forming or of what is on our approaches now. Showcase a software demo that turns many disagreeing inputs into one picture somebody can act on.

§2

THINGS TO TAKE NOTE OF

01

STRATEGIC WARNING AND TACTICAL APPROACH

Weeks to months: is something building — force movements, logistics patterns, traffic ceasing to behave normally, language shifting in step across nominally independent channels. Minutes to hours: what is on the approach now. These are treated as two professions with two toolsets. They are closer to one problem sampled at different rates, and almost nobody builds anything spanning them.

02

DATA ASSOCIATION WITHOUT SHARED IDENTIFIERS

No two vendors number things alike, so the determination is made from geometry and motion alone. It degrades worst when targets are close together — which is to say in a swarm, which is to say when it is needed.

03

TEMPORAL ALIGNMENT ACROSS SENSORS

A camera at sixty frames a second and a radar sweeping twice a second never see the same moment. Fusing them means inferring the interval, and the inference is least reliable under manoeuvre. Add clock drift, network delay and timestamps applied on receipt.

04

PER-SOURCE UNCERTAINTY MODELLING

A long-range detection and a precise one may be hundreds of metres apart and remain the same object. Unmodelled, a confident, precise, wrong sensor will dominate the picture.

05

MANIPULABLE AND ADVERSARIAL SOURCES

Anticipation runs on material nobody controls and anybody can manipulate. A warning that cannot state which of its inputs would have to be false for it to be wrong is not usable for anything that matters.

06

LABELLED DATA SCARCITY

Multimodal fusion is bounded by scarce labelled datasets. Nobody trains a multimodal foundation model over a weekend and no labelled defence corpus is waiting. Compose rather than train: take pretrained representations that exist and build the fusion layer on top.

07

FROM PICTURE TO TASKING

The path from a merged display to somebody tasking something is where these systems die, and it is almost never the part that gets built.

§3

TWO EXAMPLES

ONE — SEEING IT COMING, FROM OPEN SOURCES

VIEWS, run by a Norwegian peace research institute and a Swedish university, publishes monthly machine-learning forecasts of armed conflict one to thirty-six months ahead worldwide, and identified seven of the ten deadliest countries in 2024. It can be downloaded. Separately, people posting online behave as a distributed sensor network: untrained people photograph what they should not, and official phrasing moving in lockstep across nominally independent channels indicates an order was issued. Automated tools geolocate video from terrain and shadow angles to within a few metres. The open ground is between those two scales, and nothing joins them.

TWO — A BRAIN THAT INFERS ACROSS EVERYTHING

One system, modular interfaces, any input, a comprehensive picture out. Anduril's Lattice does this; NATO selected it for next-generation air command and control and the US Army has committed up to twenty billion dollars. The instructive detail is an exercise in which it linked F-16s, surface-to-air launchers, MQ-9 Reapers and Army howitzers into a single network, none of which Anduril built. The value lay entirely in integrating other people's equipment.